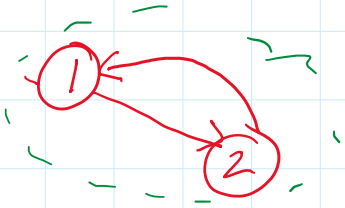
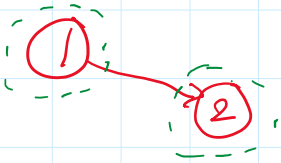
TopSort, SCC

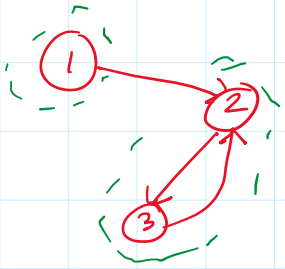
Tuesday, August 4, 2026 9:49 PM



(1-2) ✓
(2-1) ✓ } SCC_1



(1-2) ✓ } ① → SCC_1
(2-1) X } ② → SCC_2



(2-3) ✓
(3-2) ✓

(2,3) → SCC_1
(1) → SCC_2

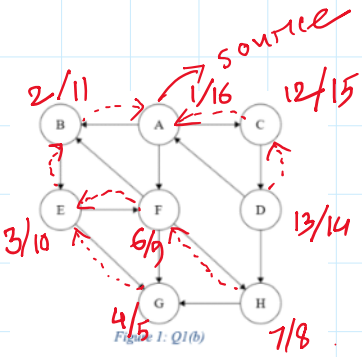


Figure 1: Q1(b)

Kosaraju's Algorithm

① Run DFS on the given graph and record end times

visited

A	B	C	D	E	F	G	H
1	1	1	1	1	1	1	1

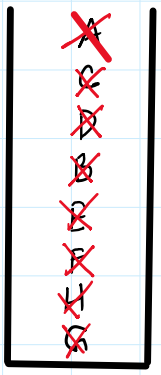
② Sort the end times in descending order

16	15	14	11	10	9	8	5
A	C	D	B	E	F	H	G

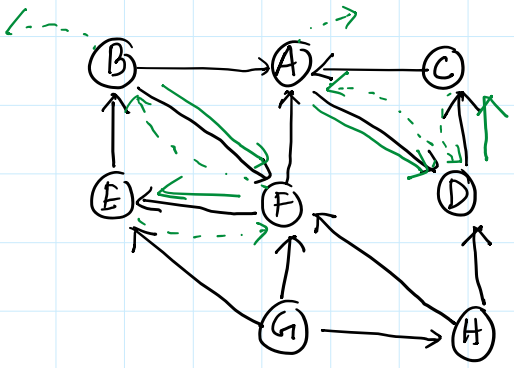
③ Push the vertices in the descending order in a stack

~~A~~

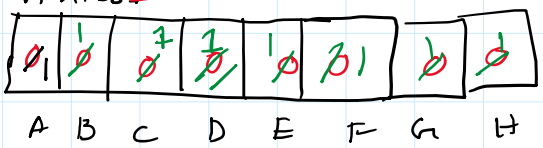
④ Transpose the graph and Run DFS according to



④ Transpose the graph and Run DFS according to Reference stack



visited



DFS(A) → DFS(D) → DFS(C)

{A, D, C} : SCC1

DFS(B) → DFS(F) → DFS(B)

{B, F, E} : SCC2

DFS(H) → x

{H} → SCC3

DFS(G) → x

{G} → SCC4